United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Final Exam, Fall 2024, MSCSE Program

Course: CSE 6213 Natural Language Processing, Marks: 40

There are SIX Questions. Answer any FOUR. Numerical figures in the right margin indicate full marks.

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1. a) Find a regular expression that supports the following English Sentences. [2]
   b) Find a grammar for the regular expression of Ques 1(a). [2]
   c) Write down the top-down and bottom-up parse trees for "He does not eat rice" [2]
   d) Find Left and Right Most Derivation for the English sentence "He does not eat rice". [2]
   e) Using the shift reduce parser check whether the given test sentence is supported by the grammar of Ques 1(b). [2]

   We eat rice
   I eat rice
   He eats rice
   Rahim eats meat
   I eat mango
   They eat mango
   I do not eat fish
   He does not eat banana

2. a) Define language model? Why is language model probability needed? [2]
   b) What are the problems occurred during language model probabilities' calculation? How can it be resolved? Explain with English Language examples. [2]
   c) Calculate unigram probability P(They eat rice) from the following English Corpus. [2]

   We eat rice
   I eat rice
   He eats rice
   Rahim eats rice
   I eat mango
   They eat mango

   d) Calculate bigram and trigram probabilities of "We eat rice" using the corpus of Ques. 2 (c). [2]
   e) For avoiding the sentence "The cow flies in the sky" which language model is necessary? How can that model solve the problem? [2]

3. a) Using the Bag of Words method calculate the input vector for the English Sentence "I do not eat rice but eat fish" using the count and frequency methods. Here the language corpora is given below. [3]

   I eat rice
   They eat rice
   He does not eat rice
   We do not eat fish
   I eat fish but not rice

b)Using the TF-IDF method, check whether the given word “example” is signifant or not for the Document 1 and Document 2. [3]

Document 1

Term	Term Count
this	1
is	1
a	2
sample	1

Document 2

Term	Term Count
this	1
is	1
another	2
example	3

c) Using the Hash Trick insert the words “RAT”, and “MAT” in the hash table of size 4. [2]

d) Give an idea to reduce the input vector size in Bag of Words method which is required for the machine learning tool(s). Explain with examples. [2]

4. a) Show three examples for each of the following affixations in Bangla Language [2]
   Prefix, Infix, Suffix
   b) Give three Bangla examples for each of inflectional and derivational morphemes. [2]
   c) Use three Bangla examples for each of Root and Stem used in morphology. [2]
   d) Write three Bangla examples for each of the free and bound morphemes. [2]
   e) What are the examples of Clipping, Coining and Acronyms in Bangla Language. [2]

5. a) Transcribe the following English words using IPA (International Phonetic Alphabet) [2]
   guarantee, ringtone, education, dogs, history
   b) Show examples for orthographic and phonetic transcription. [2]
   c) Draw a multi-class confusion matrix for parts of speech (POS) tagging. Assume random data inside the matrix [2]
   d) Find Precision, Recall, and F1-Score for each class of 5© [2]
   e) Calculate weighted average Precision, Recall, and F1-Score from 5(d) [2]

6. a) Using the Transfer Approach find the translation in English for the Bangla Sentence "বালকেরা মাঠে ক্রিকেট খেলিতেছে". Assume your own look up table. [2]

b) Find a target language sentence in Bangla for the given source language in English using the corpus-based approach. [2]

Source sentence-I: They are playing
Target sentence-II: তারা খেলিতেছি

Source sentence-II: Farmers are working in the paddy field
Target sentence-II: কৃষকেরা ধান ক্ষেতে কাজ করিতেছে

Source sentence-III: Boys are playing cricket in the field
Target sentence-III: বালকেরা মাঠে ক্রিকেট খেলিতেছে

Source sentence-IV: They repair football in the shop
Target sentence-IV: তারা দোকানে ফুটবল মেরামত করে

Source sentence-V: They buy paddy from the shop
Target sentence-V: তারা দোকান থেকে ধান ক্রয় করে।

Source language sentence: They are playing cricket in the paddy field
Target language sentence: ???

c) Determine similarity indices between the following two English documents using i) the Cosine Similarity Measure and ii) Jaccard Similarity Measure [4]

Document-I: Our spoken language is Bangla. Many people sacrifice their lives for the language.
Document-II: Bangla is our language from heart. In language movement many people participate.

d) Express through IPA (International Phonetic Alphabet) [2]
ক্রয়, দোকান, ফুটবল, মেরামত